CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Fifth Semester of B.Pharm. Examination
University Theory Examination Nov 2016
MA331 Bio-Statistics

Date: 30/11/2016 Wednesday **Time: 10:00 a.m. to 01:00 p.m.** **Maximum Marks: 80**

Instructions:

(1) **Figures to right indicate marks.**
(2) **Section wise separate answer book to be used.**
(3) **Draw neat sketches wherever necessary.**
(4) **Use of scientific non-programable calculator is allowed.**

SECTION I

Q.1. Attempt any TWO of the following.

(a) Define the following terms. [10]

(i) Arithmetic Mean
(ii) Median
(iii) Geometric Mean
(iv) Standard deviation
(v) Mean deviation

(b) Write with explanations whether following statements are true or false. [10]

(i) If the mean of 5 values of variable x is 8, and $\sum_{i=1}^{4} x_i = 20$, then $x_5 = 20$.

(ii) The standard deviation of values $x_1, x_2, \ldots x_4$ and standard deviation of values $-x_1, -x_2, -x_3, -x_4$ are equal

(iii) The median of data 54,52,56,55,55 is 56

(iv) Consider the data in table 1

Table 1

Group I	Group II
$\bar{x}_1 = 18$	$\bar{x}_2 = 12$
$n_1 = 4$	$n_2 = 6$

The arithmetic mean $\bar{x}$ of combined group is 15

(v) The geometric mean of values x_1, x_2, x_3, x_4 and geometric mean of values $2x_1, 2x_2, 2x_3, 2x_4$ are equal

(c) Write mathematical properties of [10]

(i) Arithmetic Mean (ii) Geometric Mean (iii) Standard deviation

Q.2. Attempt any <u>TWO</u> of the following.

(a) The data in Table 5 shows 44 results, representing measurements of the glucose content of routine test samples. The data are presented in the order of measurement; the units are g per 100 mL

Table 2: measurements of glucose content (g per 100 mL) of orange juices

2.27	2.30	2.23	2.27	2.27	2.30	2.30	2.26	2.34	2.34
2.50	2.46	2.12	2.09	2.16	2.13	2.18	2.28	2.28	2.42
2.43	2.53	2.56	2.55	2.50	2.60	2.52	2.62	2.64	2.57
2.56	3.03	2.20	2.94	2.99	1.95	1.86	3.07	2.15	2.48
2.21	2.26	1.47	1.52						

Construct

(i) Frequency distribution using class limits
$1.45-1.76, 1.77-2.08, \ldots, 3.05-3.36$ (03)

(ii) Relative frequency distribution and (03)

(iii) Histogram (04)

(b) The data in Table 3 shows raw calcium, milligrams per deciliter of 8 subjects 65 years of age and older.Compute [10]

Table 3: laboratory values of elderly subjects

Subjects	1	2	3	4	5	6	7	8
CAMMOL	2.53	2.50	2.43	2.48	2.33	2.13	2.55	2.45

(i) Arithmetic mean and standard deviation (05)

(ii) Mean deviation about mean . (05)

(c) The data in Table 4 shows the percentage of paracetamol in batches of Sterwin 500 mg tablets. The tablets were assayed by BP1993 UV method.Compute [10]

Table 4: Paracetamol content (%) in batches of tablets

Batch No	1	2	3	4	5	6	7
UV	84.06	84.10	85.13	84.96	84.96	84.87	84.75

(i) All quartiles (ii) Mean square deviation about 84%.

SECTION II

Q.3. Attempt any <u>TWO</u> of the following.

(a) Suppose that total carbohydrate intake in 12- to 14-year- old boys is normally distributed, with mean = 124 g/1000 cal and standard deviation = 20 g/1000 cal.What percentage of boys in this age range have carbohydrate intake [10]
(i) above 140 g/1000 cal? (ii) below 90 g/1000 cal? (iii) between 90 g/1000 cal and 140 g/1000?
You are given $P(Z < 0.8) = 0.7881, P(Z < 1.7) = 0.9554$

(b) Suppose the number of deaths from typhoid fever over a 1-year period is Poisson distributed with parameter $\mu = 4.6$. What is the probability distribution of the number of deaths over a 6-month period? **[10]**

(c) Suppose 20% of grade-school students nationwide develop influenza.what is the probability of obtaining
(i) at least 6 cases (ii) exactly 6 cases
of 15 students in grade-school class develop influenza if the nationwide rate holds true? Use Binomial distribution. **[10]**

Q.4. Attempt any <u>TWO</u> of the following.

(a) People with diabetes must manage their blood sugar levels carefully. They measure their fasting plasma glucose (FPG) several times a day with a glucose meter. Another measurement, made at regular medical checkups, is called HbA. This is roughly the percent of red blood cells that have a glucose molecule attached. It measures average exposure to glucose over a period of several months. Table 5 gives data on both HbA and FPG for 18 diabetics five months after they had completed a diabetes education class. **[10]**

Table 5: Two measures of glucose level in diabetics

Subject	HbA (%)	FPG (mg/ml)	Subject	HbA (%)	FPG (mg/ml)	Subject	HbA (%)	FPG (mg/ml)
1	6.1	141	7	7.5	96	13	10.6	103
2	6.3	158	8	7.7	78	14	10.7	172
3	6.4	112	9	7.9	148	15	10.7	359
4	6.8	153	10	8.7	172	16	11.2	145
5	7.0	134	11	9.4	200	17	13.7	147
6	7.1	95	12	10.4	271	18	19.3	255

(i) Make a scatterplot of the data.

(ii) Is the association between these variables positive or negative? What is the form of the relationship? How strong is the relationship?

(b) Consider the experiment on determination of antibacterial drug residues in catfish muscles. The % recoveries of two drugs, which were added to the muscle at approximately 20 μg g^{-1} levels, are shown in Table 6. If μ denotes the true mean (%) recovery then determine 95% and 99% confidence interval for μ. [10]

Table 6: Recoveries of drug residues in catfish muscles (%)

Oxolinic acid	95.4	97.0	96.5	103.0	94.9	95.9

You are given $t_{5,0.05/2} = 2.571, t_{5,0.05} = 2.015, t_{5,0.01/2} = 4.032, t_{5,0.01} = 3.365$

(c) The chemical of compound y, which were dissolved in 100 grams in water at various temperatures, x were recorded as follows. [10]

Table 7: Data for Regression

$x(^oc)$	15	15	30	30	45	45	60	60
y (grams)	12	10	25	21	31	33	44	39

(i) Construct the scatter plot of variables in table (7)

(ii) Find the line of regression of y on x and estimate the amount of chemical that will dissolve in 100 grams of water at 50 ^0c.